

**1. Name of the package, and what is the basic aim of what the package does or solve?**

PySpecKit (v1.0.4) is a toolkit and library used for spectroscopic analysis. The package provides features such as baseline fitting, model fitting, a measurements toolkit, and more. As well as multiple optimizations for astronomical analysis. It uses an object class called "Spectrum", which can read and fit data files for many different spectra types.

**2. Why/how did you select this package?**

I'm in another class (PHYS 456, Making Physics Experiments) where we work on a group project for the majority of the semester, and my group essentially made a crude spectrometer, so this package piqued my interest because of my other work in the area of spectral analysis.

**3. how old is the package? does it have a geneology, i.e. what related codes came before or after. are there other codes that solve the same problem?**

From what I could find, the first version of this package was released March 12, 2012. Version 1.0 was released on April 11, 2022. There doesn't seem to be a geneology for this code. There are other codes that solve the same problem, a couple I could find were SpectraFit and Specutils.

**4. is it still maintained, and by the original author(s)?**

It is still maintained, with occasional updates posted to github. One of the two original authors, Adam Ginsburg, is still the main contributor, while the other author, Jordan Mirocha, doesn't seem to be working on it anymore. The author also allows users to contribute to bug fixes.

**5. evaluate how easy it was to install and use. What commands did you use to install?**

Installing the latest version of Pyspeckit was quite simple. Using Jupyter Notebook, I just used “!pip install pyspeckit”, then to check the version I used “print(pyspeckit.\_\_version\_\_)”. For this package astropy, matplotlib, and numpy are required as well, which can be installed in a similar way.

**6. does it install via the "standard" pip/conda, or is it more complex?**

I used the standard pip method, but there are ways to install through git and The method may also change depending on operating system, but this was using Windows. You can also download the files using their pypi.org page.

**7. is the source code available? For example, "pip install galpy" may get it to you, but where can you inspect the code?**

The source code is all available to inspect on the Github. The Github also includes instructions for licensing, how to use the package, example data/code, etc.

**8. is the code used by other packages (if so, give one or two examples)**

I could not find any other Python packages that require pyspeckit.

**9. give an example how you use the code. Is it commandline, or jupyter notebook, or a web interface?**

I used the code using Jupyter Notebook to read the ngc6503vopt.fits file, since I was most familiar with that, plus installation using the same software was simple. I do not believe you can use a web interface to use Pyspeckit.

**10. provide examples using the code. if you prefer to use a jupyter notebook instead of a python script, that's ok.**

I used Jupyter Notebook since that is what I'm most familiar with. An example of some code used is

```
sp = pyspeckit.Spectrum(data=data, error=error, xarr=xaxis, xarrkwargs={'unit': 'km/s'},
unit='erg/s/cm^2/AA');sp.plotter()
```

This code uses the Spectrum class to analyze spectral data, in this case manufactured data.

Then `sp.plotter()` is used to plot said data, similar to `ax.plot()`.

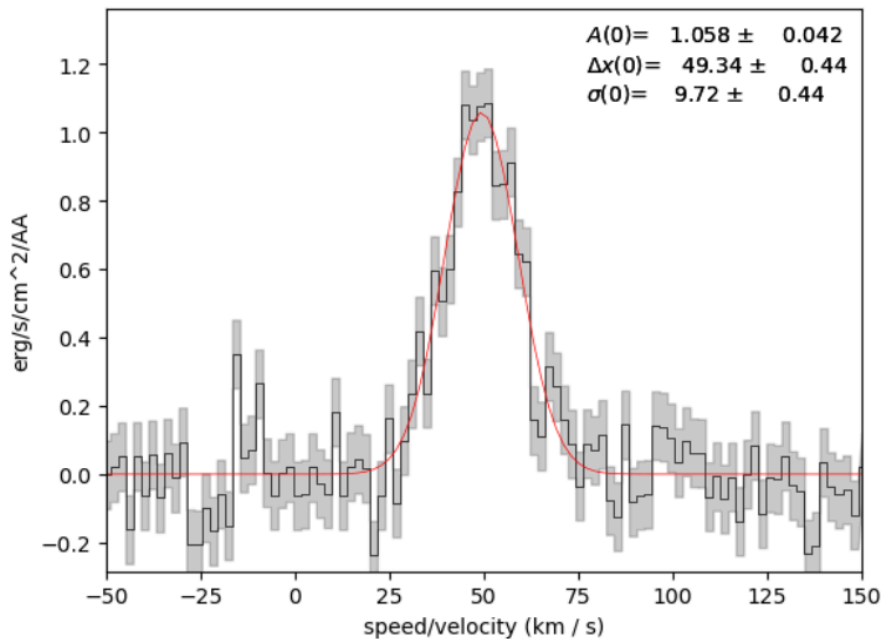
**11. does the package produce figures, or are you on your own? Is matplotlib used?**

The package produces figures. The example code I used didn't import matplotlib, however the developers stated matplotlib is required. That on top of the plot looking exactly like the matplotlib style leads me to believe matplotlib is still used somewhere in the background of this package to help on the visual end.

**12. your code and report should show at least one figure, and create a nice figure caption explaining what it shows. Your notebook should show how the figure was made (reproducible)**

I tried a couple different things, all using Jupyter Notebook. First I tried using one of their examples in the Pyspeckit Github but it looks like most of those are outdated code and might be why I couldn't get them to work. However, I found some more example code on their development website (<https://pyspeckit.readthedocs.io/en/latest/>) that I was able to modify and get to work. This example code uses synthetic data but showcases the data analysis features of Pyspeckit.

WARNING: No header given. Creating an empty one.



This example utilizes “specfit” which fits the spectrum to a specific type, in this case a gaussian fit type. I also tried plotting the ngc6503vopt.fits file with the help of online sources, but I’m not sure if it was entirely successful. I included this code in cell 2 of my notebook.

### **13. is the package pure python? or does it need accompanying C/C++/Fortran code?**

The package is pure Python, and is intended as such. All of the documentation and examples of implementation use Python. In Github you can also see the percentages of a projects’ programming languages. This one says 100% Python.

### **14. what is the input to the package? Just parameters, or dataset(s), or can they be generated from scratch?**

While the main appeal of the package is the ability to read and write datasets of multiple file types such as plain text, FITS, hdf5, and more, input can also be generated from scratch and analyzed just the same as external data. The package’s website also advertises very flexible model parameters.

### **15. what is the output of the package? Just parameters, or dataset(s)?**

The output of the package is normally a plot with a spectral fit.

**16. does the code provide any unit tests, regression or benchmarking?**

In the Github there is a pyspeckit-tests file which provides some test code as well as benchmarking updates.

**17. how can you feel confident the code produce a reliable result? (see also previous question)**

Along with seeing the mentioned benchmarks and tests, one other method to test the reliability of the code is to use a dataset of known spectrum to test the accuracy based on what you know the true output to be.

**18. what (main) python package(s) does it use or depend on (e.g. numpy, curve\_fit, solve\_ivp) - how did you find this out?**

The README file contains lots of useful information, including required packages. It says astropy, matplotlib, and numpy are required, while scipy, lmfit, lineid\_plot, and spectral-cube are optional.

**19. what kind of documentation does the package provide? was it sufficient for you?**

Luckily, this package provides lots of documentation written over the entire span of this package's life. Especially on the dedicated pyspeckit website. For the most part, this documentation was sufficient for me to debug and generally learn more about the package. One frustrating part though was that while most of the documentation was still up to date, many of the code examples were not up to date with current versions of python and/or the required packages.

**20. if you use this code in a paper, do they give a preferred citation method?**

Yes, there is a preferred citation method. They provide a link to the Harvard ADS, which has a page for exporting a citation in many different formats.

**21. provide any other references you used in your report.**

Astropy: <https://ascl.net/1304.002>

**22. can you find two other papers that used this package. E.g. use ADS citations for ASCL based code.**

Using ADS citations I found a reference in “Physical Conditions of the Ionized Superwind in NGC 253 with VLT/MUSE” (<https://ui.adsabs.harvard.edu/abs/2025arXiv250504707C/abstract> ), as well as “Galactic H<sub>2</sub>CO Densitometry. I. Pilot Survey of Ultracompact H II Regions and Methodology” (<https://ui.adsabs.harvard.edu/abs/2011ApJ...736..149G/abstract> ). Similar to NEMO’s issue, Pyspeckit seems to have two different pages in the ADS.

**23. did you have to learn new python methods to use this package?**

Not a lot, luckily we already used all of the required packages in class, however using the new Spectrum class and some more of the custom tools had a learning curve. As mentioned before the detailed documentation helped greatly.

**24. Final Disclaimer: you need to state if you have prior experience in using the package or the data, or this is all new to you. In addition, if you collaborated in a group, as long as this is your work.**

I had no prior experience using this package, or any astronomical data. I also did not collaborate in any group.